

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Winter Examination – 2024****Course: B.Tech****Branch :Petrochemical Engineering****Semester : VI****Subject Code & Name: BTPCC602 Mass transfer -II****Max Marks: 60****Date:20/12/2024****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Which one of the following is a constant reflux process? a. Flash distillation b. Simple distillation c. Multicomponent distillation d. Final distillation		1
2	When the mixture to be distilled has a very high boiling point and the product material is heat sensitive, the separation technique to be used is _____ distillation. a. Continuous b. Steam c. Azeotropic d. None of these		1
3	If $F = 200$ kmol/h and $z_F = 0.4$, $x_D = 0.95$, $x_W = 0.04$ and $R = 2.0$, Then D is a. 88.1 b. 22.6 c. 79.1 d. 158.2		1
4	Henry's law states that the a. Partial pressure of a component over a solution is proportional to its mole fraction in the liquid b. Partial pressure of a component over a solution is proportional to the mole fraction in the vapour c. Vapour pressure is equal to the product of the mole fraction and total pressure d. D. Partial pressure is equal to the product of the mole fraction and total pressure		1
5	Drift in a cooling tower is a. The water entrained by the circulating air b. Dependent on the water lost by evaporation c. Desirable d. All of the above		1

6	Dry bulb temperature of the gas is _____ the wet bulb temperature. a. Less than b. More than c. Equal to d. None of these		1
7	Advantage of continuous drying over batch drying is that the a. Drying cost per unit weight of dried product is less b. Product with more uniform moisture content is produced c. Equipment size required is smaller d. All of the above		1
8	The drying time between fixed moisture contents within the 'constant rate period' is proportional to (assuming that drying occurs from all surfaces of the solid) (where, T = thickness of the solid) a. T^2 b. T c. $T^{1.5}$ d. T^3		1
9	In cooling tower 'Approach ' is the temperature difference between a. DBT and WBT of air b. cold water outlet and WBT of air c. hot water inlet and WBT of air d. hot water inlet and cold water outlet		1
10	Which of the following is the example of crystallization process? a. Purification of alum b. Purification of sea water c. Separation of gases from air d. None of the mentioned		1
11	Which of the following mechanisms is not used by a membrane filter to remove particulate material from the water? a. Flux b. Reverse flow c. Particle counts d. Chlorine		1
12	Why is osmosis not a good separation process? a. Because it involves semipermeable membrane b. Because it transfers solvent from wrong direction c. Because it involves movement of solute particles d. Because it requires high temperatures		1
Q. 2	Solve the following.		12
A)	Define distillation with neat flow diagram. Give the methods or classification of distillation.	remembering	6
B)	Explain the following terms a) Vapor Liquid Equillibria b) Relative volatility c) Bubble point dew point	remembering	6

Q.3	Solve the following.		12
A)	Determine total number of stages and feed stage for separation of a binary mixture with relative volatility (α) 2.5, Feed composition (z_F) 0.36, Distillate Composition (x_D) 0.915, Bottom composition (x_B) 0.05 and feed is saturated liquid. Reflux ratio to be 1.55.	applying	6
B)	Tabulate different condition of feed along with their q values. Represent q lines for different feed conditions on equilibrium diagram.	Remembering	6
Q. 4	Solve Any Two of the following.		12
A)	Explain Adiabatic saturation of air in detail.	Understanding	6
B)	Write a short note on following: a) Psychrometric chart b) Dry bulb temperature and Wet bulb temperature	Remembrance	6
C)	How the cooling towers are classified? Describe any one type of cooling tower with neat diagram	Understanding	6
Q.5	Solve Any Two of the following.		12
A)	Draw the drying rate curve with nomenclature. A wet solid of 150 kg is dried from a moisture content of 45 wt % to 10 wt%. The critical moisture content is 15 wt % and equilibrium moisture content is zero. All above quantities are on dry basis. The falling rate is assumed to be zero and it takes 5 minutes to dry material in constant rate period. Find the total duration of drying.	Analyzing and applying	6
B)	With neat sketch, explain construction and working of (i) tray dryer (ii) rotary dryer	Remembrance and understanding	6
C)	Write short note on (a) Primary nucleation (b) Law of crystal growth (c) Comparison of batch crystallizer and continuous crystallizer	Remembrance	6
Q. 6	Solve Any Two of the following.		12
A)	Classify the different membranes separation processes based on different driving force and separation size range. What are the specific examples for the same?	Understanding	6
B)	Write down the types of fouling in micro-filtration.	Remembrance	6
C)	Why nano-filtration is called as ultra-osmosis? Describe the Industrial application of Nano-filtration	Remembrance and understanding	6
	*** End ***		